

Functions of Several Variables & Partial Derivatives

ANSWER KEY



Evaluating functions of two variables

- 1 For $f(x, y) = x^2y - 3xy + y^2$, find $f(2, -1)$.
 $f(2, -1) = (4)(-1) - 3(2)(-1) + 1 = -4 + 6 + 1 = 3$.
- 2 For $f(x, y) = \sqrt{x^2 + y^2}$, find $f(3, 4)$ and describe the domain of f .
 $f(3, 4) = \sqrt{9 + 16} = 5$. Domain: all of $\mathbb{R}^2$, since $x^2 + y^2 \geq 0$ always.
- 3 For $g(x, y, z) = xy + yz + zx$, find $g(1, 2, 3)$.
 $g(1, 2, 3) = 1(2) + 2(3) + 3(1) = 2 + 6 + 3 = 11$.
- 4 For $f(x, y) = \frac{x-y}{1+y}$, find $f(5, 3)$ and state one point where f is undefined.
 $f(5, 3) = \frac{2}{4} = \frac{1}{2}$. f is undefined whenever $x + y = 0$, e.g., at $(1, -1)$.

Computing partial derivatives

- 5 For $f(x, y) = x^4y^3 + 2xy - y^5$, find f_x and f_y .
 $f_x = 4x^3y^3 + 2y$. $f_y = 3x^4y^2 + 2x - 5y^4$.
- 6 For $f(x, y) = e^{x^2y}$, find f_x and f_y .
 $f_x = 2xy e^{x^2y}$. $f_y = x^2 e^{x^2y}$.
- 7 For $f(x, y) = \ln(x^2 + y^2)$, find f_x and f_y .
 $f_x = \frac{2x}{x^2 + y^2}$. $f_y = \frac{2y}{x^2 + y^2}$.
- 8 For $f(x, y) = x \sin(xy)$, find f_x and f_y .
By the product rule, $f_x = \sin(xy) + x \cdot y \cos(xy) = \sin(xy) + xy \cos(xy)$. $f_y = x \cdot x \cos(xy) = x^2 \cos(xy)$.

Second-order and mixed partial derivatives

- 9 For $f(x, y) = x^3y^2$, find f_{xx} , f_{yy} , and f_{xy} .
 $f_x = 3x^2y^2$, $f_y = 2x^3y$. $f_{xx} = 6xy^2$. $f_{yy} = 2x^3$. $f_{xy} = 6x^2y$.
- 10 For $f(x, y) = e^{xy}$, find f_{xx} and f_{xy} .
 $f_x = ye^{xy}$. $f_{xx} = y^2e^{xy}$. $f_{xy} = e^{xy} + xy e^{xy} = (1 + xy)e^{xy}$.
- 11 For $f(x, y) = \sin(x)\cos(y)$, find f_{xy} .
 $f_x = \cos(x)\cos(y)$. $f_{xy} = -\cos(x)\sin(y)$.

Clairaut's Theorem: equality of mixed partials

- 12** For $f(x, y) = \sin(xy) + x^2y$, verify that $f_{xy} = f_{yx}$.
 $f_x = y\cos(xy) + 2xy$, so $f_{xy} = \cos(xy) - xy\sin(xy) + 2x$. $f_y = x\cos(xy) + x^2$, so $f_{yx} = \cos(xy) - xy\sin(xy) + 2x$. Equal, as expected.
- 13** State Clairaut's Theorem and the condition it requires.
If f_{xy} and f_{yx} are both continuous on an open region, then $f_{xy} = f_{yx}$ throughout that region.
- 14** For $f(x, y) = x^3y^4$, compute f_{xy} and f_{yx} separately to confirm Clairaut's Theorem.
 $f_x = 3x^2y^4 \Rightarrow f_{xy} = 12x^2y^3$. $f_y = 4x^3y^3 \Rightarrow f_{yx} = 12x^2y^3$. Equal.
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Domains and level curves

- 15** Find the domain of $f(x, y) = \ln(x - y)$.
Requires $x - y > 0$, i.e., $x > y$. Domain: $\{(x, y) : x > y\}$.
- 16** Describe the level curve $f(x, y) = 4$ for $f(x, y) = x^2 + y^2$.
 $x^2 + y^2 = 4$ is a circle of radius 2 centered at the origin.
- 17** Find the domain of $f(x, y) = \sqrt{9 - x^2 - y^2}$ and describe it geometrically.
Requires $9 - x^2 - y^2 \geq 0$, i.e., $x^2 + y^2 \leq 9$. Domain: the closed disk of radius 3 centered at the origin.
- 18** Describe the level curves $f(x, y) = c$ for $f(x, y) = y - x^2$, for a general constant c .
 $y - x^2 = c \Rightarrow y = x^2 + c$: each level curve is an upward parabola, shifted vertically by c .
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Functions of three variables

- 19** For $f(x, y, z) = x^2 + y^2 + z^2$, find f_x , f_y , f_z .
 $f_x = 2x$, $f_y = 2y$, $f_z = 2z$.
- 20** Describe the level surface $f(x, y, z) = 9$ for $f(x, y, z) = x^2 + y^2 + z^2$.
A sphere of radius 3 centered at the origin.
- 21** For $w = x^2yz^3$, find w_x , w_y , and w_z .
 $w_x = 2xyz^3$. $w_y = x^2z^3$. $w_z = 3x^2yz^2$.
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Interpreting partial derivatives

- 22 The temperature at a point on a metal plate is $T(x, y) = 50 - 0.1x^2 - 0.2y^2$. Compute $T_x(2, 3)$ and interpret its sign.
 $T_x = -0.2x$, so $T_x(2, 3) = -0.4$. Since this is negative, moving in the positive x -direction from $(2, 3)$ while holding y fixed causes the temperature to decrease, at an instantaneous rate of 0.4 degrees per unit distance.
- 23 For the production function $P(L, K) = 50L^{0.6}K^{0.4}$ (Cobb–Douglas, L =labor, K =capital), state in words what P_L represents.
 P_L is the marginal product of labor: the instantaneous rate of change of output P per additional unit of labor L , holding capital K fixed.

Higher-order partial derivatives with three variables

- 24 For $f(x, y, z) = x^2yz + y^3z^2$, find f_{xz} .
 $f_x = 2xyz$, so $f_{xz} = 2xy$.
- 25 For $f(x, y, z) = e^{xyz}$, find f_x , then f_{xy} .
 $f_x = yze^{xyz}$. $f_{xy} = ze^{xyz} + yz \cdot xze^{xyz} = z(1 + xyz)e^{xyz}$.

Sketching and identifying surfaces from level curves

- 26 The level curves of $f(x, y) = x^2 - y^2$ at $c = 1, -1, 0$ are hyperbolas (and a pair of crossing lines at $c = 0$). Find the equations of the level curves for $c = 4$ and $c = -4$.
 $c = 4$: $x^2 - y^2 = 4$, a hyperbola opening left-right. $c = -4$: $x^2 - y^2 = -4$, i.e., $y^2 - x^2 = 4$, a hyperbola opening up-down.
- 27 For $f(x, y) = \frac{1}{x^2 + y^2 + 1}$, describe the level curve $f(x, y) = \frac{1}{2}$.
 $\frac{1}{x^2 + y^2 + 1} = \frac{1}{2} \Rightarrow x^2 + y^2 + 1 = 2 \Rightarrow x^2 + y^2 = 1$ — a circle of radius 1 centered at the origin.
- 28 Explain why the level curves of any function $f(x, y) = g(x^2 + y^2)$, for a one-variable function g with $g' \neq 0$ everywhere, are always circles centered at the origin.
Setting $f(x, y) = c$ gives $g(x^2 + y^2) = c$. Since g is one-to-one where $g' \neq 0$, this determines a single value $x^2 + y^2 = r^2$ for each c in g 's range, and $x^2 + y^2 = r^2$ is exactly the equation of a circle of radius r centered at the origin.